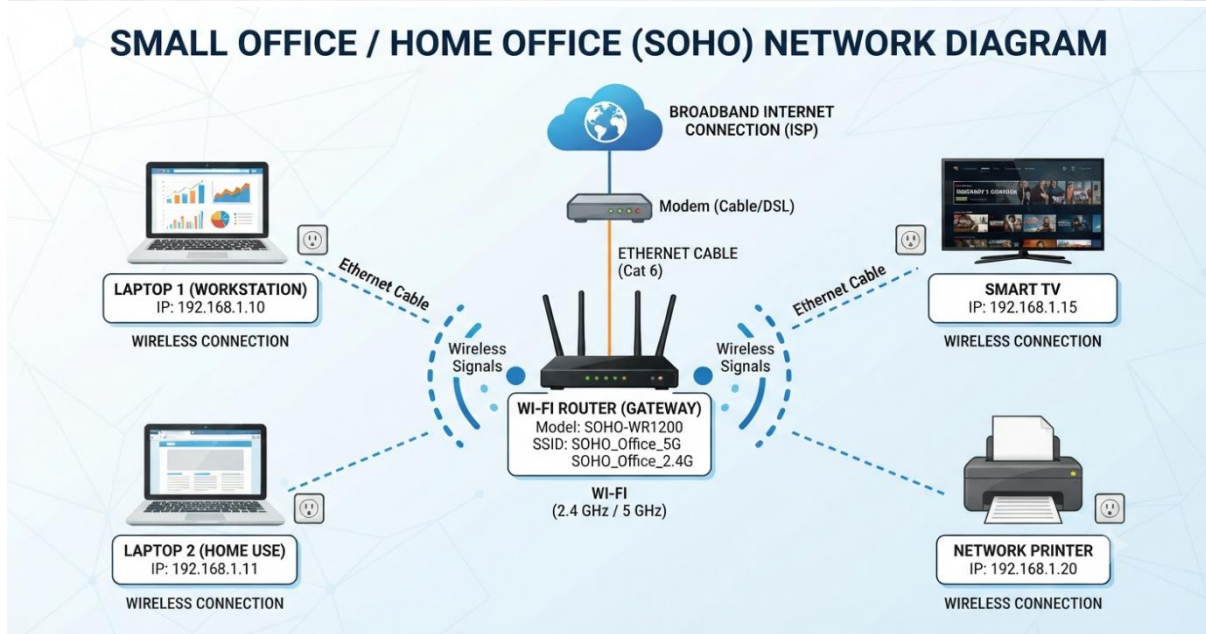
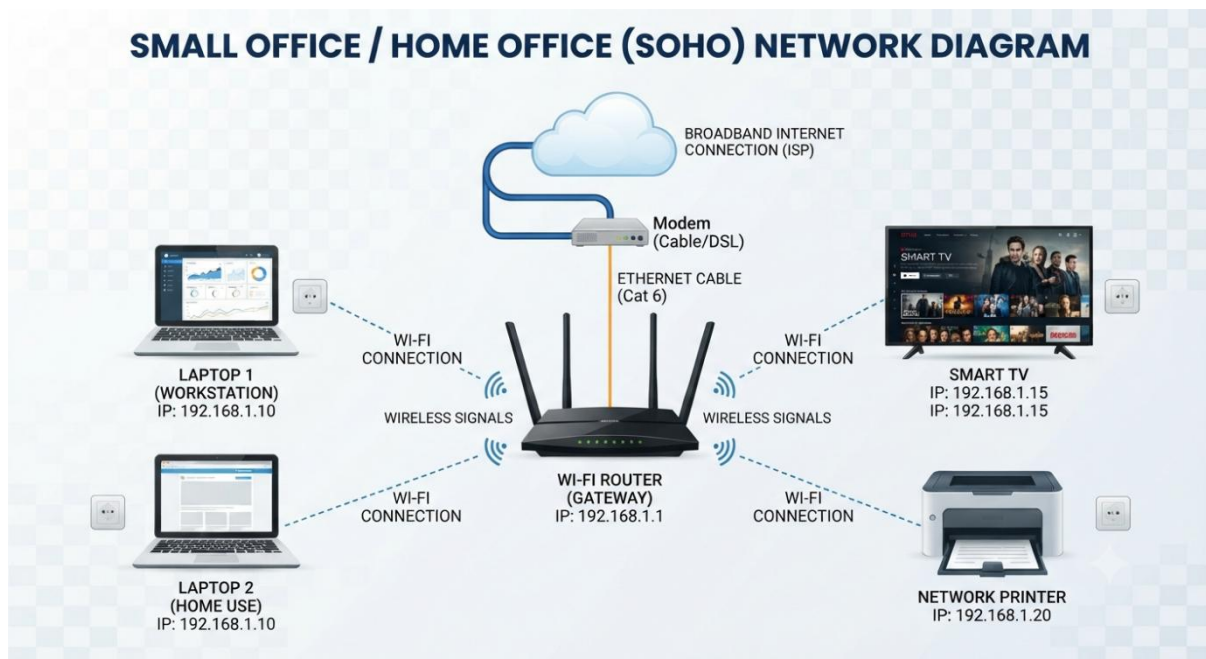


Day – 1 Build a SOHO network.

Task – 1 Diagram of small office.



Task – 2 Hardware components.

- 1. Core Network Infrastructure
- Broadband Internet Connection (ISP)
- Type: Fiber-to-the-Home (FTTH) or Cable Broadband (commonly provided via local ISPs, Airtel Xstream, or JioFiber).
- Modem / Optical Network Terminal (ONT) / Gateway
- Function: Converts incoming optical fiber or coaxial signals into Ethernet packets.
- Suggestion: TP-Link GPON/XPON ONT (e.g., XC220-G3v) or standard ISP-provided dual-band ONT/Modem unit.
- Wi-Fi Router (Gateway)
- Function: Handles internal routing, DHCP allocation, NAT, and wireless access.
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- 2. End-User and Peripheral Devices
- Workstation Laptops (Quantity: 2)
- Function: Used for daily productivity, tasks, and internet browsing.
- Smart TV (Quantity: 1)
- Function: Connected wirelessly for OTT streaming and media consumption.
- Suggestion: Samsung 43-inch Crystal 4K UHD Smart TV or Sony Bravia Smart TV with built-in Wi-Fi connectivity.
- Network Printer (Quantity: 1)
- Function: Shared printing across the SOHO network.
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- 3. Cables and Connectivity Accessories
- Ethernet Cable (Cat 6)
- Purpose: Connects the Broadband Modem/ONT to the WAN port of the Wi-Fi Router.

Task – 3 Config wifi router for SOHO network.

- Step 1: Access the Router Administration Page
- Connect a laptop to the router either via an Ethernet cable or by joining its default open Wi-Fi network.
- Open a web browser and type the router's default gateway IP address (such as 192.168.0.1 or 192.168.1.1) into the address bar.
- Log in using the default administrator credentials (typically found printed on a sticker underneath the router, such as username admin and password admin or password).
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- Step 2: Set Up a Unique SSID (Network Name)

- Navigate to the Wireless or Wi-Fi settings menu in the router's web interface.
- Locate the wireless network name fields for your frequency bands (e.g., 2.4 GHz and 5 GHz).
- Change the SSID to a unique, recognizable name for your home office (for example, SOHO_Office_2.4G and SOHO_Office_5G).
- Save the changes.
-
- Step 3: Enable WPA2 Security and Set a Strong Password
- Under the wireless settings menu, click on Wireless Security or Security Mode.
- Select WPA2-Personal (also referred to as WPA2-PSK) with AES encryption as it offers robust protection for modern home networks.
- In the Passphrase or Password field, enter a strong, unique password combining uppercase letters, numbers, and symbols.
-
- Step 4: Change the Default Administrator Password
- Navigate to the Administration, System Tools, or Management section in the router menu.
- Locate the password change settings for the router's admin account.
- Enter the current default password, followed by your new, highly secure administrator password.
- Save your settings and log out to test logging back in with your new credentials.

Task – 4 Config ip to all devices.

- Wi-Fi Router (Gateway)
- Device Type: Gateway
- MAC Address: 00:1A:2B:3C:4D:5E
- IP Assignment Type: Static (Base)
- Assigned IP Address: 192.168.1.1
- Default Gateway: N/A
-
- Laptop 1
- Device Type: Workstation
- MAC Address: B4:B5:FE:11:22:33
- IP Assignment Type: DHCP
- Assigned IP Address: 192.168.1.10
- Default Gateway: 192.168.1.1
-
- Laptop 2

- Device Type: Home Use
- MAC Address: C8:D7:19:44:55:66
- IP Assignment Type: DHCP
- Assigned IP Address: 192.168.1.11
- Default Gateway: 192.168.1.1
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- Smart TV
- Device Type: Entertainment
- MAC Address: 74:DA:38:77:88:99
- IP Assignment Type: Static
- Assigned IP Address: 192.168.1.15
- Default Gateway: 192.168.1.1
-
- Network Printer
- Device Type: Peripheral
- MAC Address: 00:22:64:AA:BB:CC
- IP Assignment Type: Static
- Assigned IP Address: 192.168.1.20
- Default Gateway: 192.168.1.1

Task – 5 Troubleshoot to laptop.

- When a laptop cannot connect to the internet in a SOHO network, it usually points to an issue with physical connectivity, local IP configuration, or gateway routing. Here are three common causes and the step-by-step diagnostic and fix procedure for each:
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- 1. Wi-Fi Association or Authentication Failure
- Cause: The laptop might be disconnected from the wireless network, experiencing weak signal strength, or failing authentication due to an incorrect Wi-Fi password (especially after a security change).
- Diagnosis: Check the network icon on the laptop's taskbar. If it shows a globe with a cross or a "No Internet, Secured" message, open your wireless settings to verify if it is actively connected to your SOHO SSID.
- Forget the wireless network on the laptop and reconnect by re-entering the correct WPA2 password.
- Move closer to the Wi-Fi router to rule out signal range issues or interference.
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- 2. DHCP Configuration or IP Address Conflict

- Cause: The laptop is set to use a static IP address that clashes with another device, or it failed to acquire an automatic IP address via DHCP from the router (resulting in a self-assigned APIPA address like 169.254.x.x).
- Change the laptop's network adapter settings to Obtain an IP address automatically (DHCP).
- In the command prompt, run `ipconfig /release` followed by `ipconfig /renew` to force the laptop to request a fresh IP from the router.
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- 3. Default Gateway or DNS Resolution Issue
- Cause: The laptop connects to the Wi-Fi and gets an IP address, but its default gateway is misconfigured, or the DNS servers are down, preventing domain names (like google.com) from resolving.
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